UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
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OFFICE OF
PREVENTION, PESTICIDES
AND TOXIC SUBSTANCES

MEMORANDUM

SUBJECT: Draft Guidance on LADDs and APDRs

FROM: Becky Brown, Industrial Hygienist Becky Brown
Chemical Engineering Branch (TS-779)

TO: CEB Staff and Management
Lois Dicker, CSRAD
Pat Kennedy, EAB
Ray Kent, CSRAD
Elizabeth Margosches, HERD

Attached are draft guidelines for estimating Lifetime Average Daily Doses (LADDs) and Acute Potential Dose Rates (APDRs) that will be used by CEB in preparing occupational exposure assessments. This guidance reflects the changes in the revised Exposure Assessment Guidelines.

Please review and comment on all aspects of these guidelines by July 7, 1993. Forward your comments to Cathy Fehrenbacher. The guidelines will be revised to incorporate comments and will then be used in CEB assessments. Future projects will include issues on breathing rates and peak exposure assessment.

Thanks for your help. Please contact Cathy (260-0696) or myself (260-0099) if you have any questions or need more information.

cc: Mary Ellen Weber
Larry Longanecker

ESTIMATING LADDs

Introduction

The Chemical Engineering Branch (CEB) estimates potential exposure to workers for new and existing chemicals.
In the past the CEB has estimated occupational exposures using potential dose rates (units in mg/day). To be consistent with exposure estimates provided by the Exposure Assessment Branch (EAB), CEB will now be calculating Lifetime Average Daily Doses (LADDs) and Acute Potential Dose Rates (APDRs). These units are in mg/kg•day. APDRs and LADDs differ only in that LADDs include factors for the number of years worked and number of years lived in a lifetime.

In the past CEB has estimated LADDs on a case-by-case basis. This document offers general guidance on estimating LADDs and APDRs. The formulas used are presented in two different ways; both as stand-alone and as the next step after calculating potential dose rates. The basic assumptions and default values generally used are also included. Estimating peak exposures is not yet well-defined, however there is some discussion of preliminary work toward defining estimation methods for peak exposures.

The format of this guidance is as follows:

- Discussion of calculations for estimating LADDs
- Attachment 1, discussion of default values
- Attachment 2, two examples from previous CEB reports
- Attachment 3, reference for breathing rates
- Attachment 4, discussion of chronic, acute and peak exposure estimation
- Attachment 5, list of references used in this guidance

ESTIMATING LADDs & APDRs

Inhalation Exposure - LADD:

To estimate the LADD for inhalation, the following equation is used (references 3 and 7).

$$LADD = \frac{C \times IR \times DU \times ED \times WT}{BW \times LT \text{ (365 days/year)}}$$

Where: LADD = Lifetime Average Daily Dose (mg/kg•day)
C = Average exposure concentration (mg/m^3)
IR = Inhalation rate (m^3/hr)
DU = Daily exposure duration (hr/day)
ED = Summary of all exposure duration for all events (days/year)
WT = Working lifetime (years)
BW = Body weight (kg)
LT = Total worker lifetime (years)

This formula can be broken down into the calculation for estimating the potential dose rate (reference 2):

$$I = C \times IR \times DU$$

Where: I = Inhalation exposure (mg/day)

and then including the weight, and working and living lifetime terms to complete the LADD calculation:

$$LADD = \frac{I \times ED \times WT}{BW \times LT \text{ (365 day/yr)}}$$

Inhalation Exposure - APDR:

Estimating the APDR for acute inhalation exposure is very similar to the LADD, without the factors for number of exposure durations per year or the total working and lifetime years.

$$APDR = \frac{C \times IR \times DU}{BW}$$

Where: APDR = Acute Potential Dose Rate (mg/kg•day)
C = peak exposure when available, otherwise average exposure (mg/m^3)
All other factors are defined above.

Dermal Exposure - LADD:

To estimate the LADD for dermal exposure, use the following equation. This equation doesn't account for the dermal absorption factor (DAF). The Health & Environmental Review Division (HERD) calculates the DAF and multiplies it with the dermal LADD or APDR to obtain the dose rate that includes absorption of the substance.

$$LADD = \frac{S \times Q \times C \times FQ \times WT}{BW \times LT\ (365\ days/year)}$$

Where: LADD = Dermal Lifetime Average Daily Dose (mg/kg•day)
S = Surface area of contact (cm^2)
Q = Quantity typically remaining on skin (mg/cm^2)
C = Fraction of chemical in mixture; also referred to as the weight fraction (WF) which is unitless
FQ = Frequency of events per year (/year)
WT = Working lifetime (years)
BW = Body weight (kg)
LT = Total worker lifetime (years)

The equation can be broken down, first using the dermal exposure equation:

$$D = S \times Q \times C$$

Where: D = Dermal exposure (mg)
S, Q and C are defined above

Then the $LADD = \frac{D \times FQ \times WT}{BW \times LT\ (365\ day/yr)}$

Dermal Exposure - APDR:

In estimating the APDR for dermal exposure, the following equation is used. Like the dermal LADD above, this equation doesn't include the DAF calculated by HERD.

$$APDR = \frac{S \times Q \times C \times E}{BW}$$

Where: APDR = Dermal Acute Potential Dose Rate (mg/kg•day)
E = Events per day (/day)
See above for definitions for all other factors.

Since D = S x Q x C, the above equation can be broken down to:

$$APDR = \frac{D \times E}{BW}$$

The total LADD or APDR consists of adding the inhalation LADD/APDR and the dermal LADD/APDR.

LADD(Total) = LADD(Inhalation) + LADD(Dermal)

APDR(Total) = APDR(Inhalation) + APDR(Dermal)

Reporting: When reporting LADDs/APDRs, include exposure descriptors, assumptions and uncertainties.

Default Values (for further discussion, see Attachment 2):

The exposure concentration is derived using standard CEB methods for the operation (reference 2).

Body weight of 70 kg (reference 1). If there are concerns of developmental toxicity, ORD recommends using 50 kg for pregnant women (reference 6), however, CSRAD and EAB do not use it.

Inhalation Rate of 1.25 m^3/hour (reference 1)

Exposure duration of 8 hours per day (default). Use the actual exposure duration if known (reference 2).

Working lifetime of 40 years (reference 1).

Total worker lifetime of 70 years (reference 2)
Total worker lifetime of 75 years (reference 1). EAD uses 75 years as a high-end estimate.

For acute exposures, the default duration of exposure is 8 hr/day.

Use of LADD/APDR in Risk Assessments:

To further define the risk, HERD uses the following equation:

$$\text{Risk} = \frac{\text{LADD x NOEL}}{\text{RF}} \quad \text{or} \quad \frac{\text{APDR x NOEL}}{\text{RF}}$$

Where: LADD is the lifetime average daily dose in mg/kg•day.
APDR is acute potential dose rate in mg/kg•day.
NOEL is the No Observable Effect Level in kg•day/mg. The LOEL or MOE can also be used.
RF is the risk factor used by HERD (unitless).

DEFAULT VALUES
AND
BASIC ASSUMPTIONS

Body Weight - 70 kg, based on average of adult women and men (reference 1). ORD recommends 50 kg for pregnant women potentially exposed to teratogenic effects (reference 6), however, EAB does not use 50 kg.

Daily Exposure Duration - default at 8 hr/day (reference 2) for both LADDs and APDRs. Use actual daily exposure duration if known.

Dermal Exposure Factors. Typical values for dermal exposure factors are found in the CEB Manual (reference 1). According to reference 1 (Exposure Factors Handbook), the surface area numbers include exposure to hands and arms. Average surface area for two hands alone is 840 cm^2 for men and 750 cm^2 for women. (NOTE: The dermal exposure method is under revision.)

Events per Day - default at 1 per day for dermal exposures (reference 2).

Exposure Concentration - based on CEB methodology (reference 2)

Inhalation Rate - 1.25 m^3/hr (reference 1). This default value represents the average adult male inhalation rate for light to moderate work and the average adult female inhalation rate for moderate work. Further discussion on breathing rates is at Attachment 3.

Summary of All Exposure Durations for All Events - default at 250 days/yr (reference 2). Use actual days per year if known.

Worker Lifetime - 70 yr/lifetime (reference 2), 75 yr/lifetime (reference 1) for high-end estimates. Both numbers are based on the average lifetime of both men and women. Reference 1 (Exposure Factors Handbook) recommends that the total lifetime be factored in only for carcinogenic risk assessments. EPA still uses 70 yr/lifetime. EAB uses 75 yr/lifetime for high-end estimates.

Working Lifetime - total number of years worked for an individual is 40 yr/lifetime (reference 1). Default value for LADDs.

EXAMPLES OF LADD ESTIMATION

Example 1 (Reference 4)
Manufacturing of Pigment Violet 23 with Chloranil
Inhalation and Dermal Exposure

Key Assumptions:

- Concentration of TCDD in Chloranil - 25 ppb
- Inhalation exposure concentration assumed to be at nuisance dust standard 15 mg/m^3 of particulate chloranil (consistent with monitoring data from chloranil processor)
- Duration of exposure is 8 hr/day
- Operating days per year is 114 based on company data
- Dermal exposure ranges up to 18,200 mg/day available for absorption (based on standard CEB assessment methodology)

Exposure Estimate:

Inhalation (Chronic Inhalation Rate)

$$\frac{15\ \text{mg}}{\text{m}^3} \times \frac{1.25\ \text{m}^3}{\text{hr}} \times \frac{8\ \text{hr}}{\text{day}} \times \frac{25\ \text{pg TCDD}}{\text{mg Chloranil}} \times \frac{\text{ng}}{1000\ \text{pg}} = 4.0\ \text{ng/day}$$

Dermal

$$\frac{18{,}200\ \text{mg}}{\text{day}} \times \frac{25\ \text{pg TCDD}}{\text{mg Chloranil}} \times \frac{\text{ng}}{1000\ \text{pg}} = 500\ \text{ng/day}$$

Risk Estimate

$$\text{LADD(i)} = \frac{4.0\ \text{ng}}{\text{day}} \times \frac{40\ \text{yr}}{\text{life}} \times \frac{114\ \text{days}}{\text{yr}} \times \frac{1}{70\ \text{kg}} \times \frac{1}{\frac{365\ \text{d}}{\text{yr}} \times \frac{70\ \text{yr}}{\text{life}}} \times$$

$$\frac{1\ \text{mg}}{1\ \text{EE6 ng}} = 1.02\ \text{EE-8 mg/kg}\bullet\text{day}$$

$$\text{LADD(d)} = \frac{500\ \text{ng}}{\text{day}} \times \frac{40\ \text{yr}}{\text{life}} \times \frac{114\ \text{days}}{\text{year}} \times \frac{1}{70\ \text{kg}} \times \frac{1}{\frac{365\ \text{d}}{\text{yr}} \times \frac{70\ \text{yr}}{\text{life}}} \times$$

$$\frac{1\ \text{mg}}{1\ \text{EE6 ng}} \times 0.012 = 1.53\ \text{EE-8 mg/kg}\bullet\text{day}$$

LADD(t) = LADD(i) + LADD(d) = 2.55 EE-8 mg/kg•day

Risk at 25 ppb TCDD contamination in chloranil

$$2.55\ \text{EE-8 mg/kg}\bullet\text{day} \times \frac{1}{0.55} \times 1.56\ \text{EE5 kg}\bullet\text{day/mg} = 7.2\ \text{EE-3}$$

Example 2 (Reference 5)
Exposure to acetone during aerosol spray painting operations
Inhalation exposure

Key Assumptions:

- Body weight of 70 kg
- Inhalation rate of 1.25 m^3/hr
- Continuous exposure during 8-hr work shift
- Exposure duration of 250 day/yr
- Working lifetime of 40 yr/life
- Worker lifetime of 70 yr/life
- 1 can of spray paint used in one 8-hr shift

$$\text{LADD} = \frac{16.56\ \text{mg}}{\text{m}^3} \times \frac{1}{70\ \text{kg}} \times \frac{1.25\ \text{m}^3}{\text{hr}} \times \frac{8\ \text{hr}}{\text{day}} \times \frac{250\ \text{day}}{\text{yr}} \times \frac{40\ \text{yr}}{\text{life}} \times \frac{1}{\frac{365\ \text{d}}{\text{yr}} \times \frac{70\ \text{yr}}{\text{life}}} = 0.926\ \text{mg/kg}\bullet\text{day}$$

Breathing Rates - Research to Date
June 8, 1993

Coal Miners

NIOSH recommendations to MSHA were to use breathing rates from a British study[1] of miners wearing respiratory protection. These recommendations are found in the NIOSH/DSDTT risk assessment for diesel exhaust[2].

Other References

Another method considered by NIOSH for the MSHA study, (but set aside when the British study was found) was to couple data on human energy expenditures[3] with calorie expenditures[4] to calculate breathing rates. The human energy expenditures book is not available through EPA's Interlibrary Loan system (6/93). Contact at NIOSH: Randy Smith (513) 533-8307.

Randy also recommended reviewing "Report on the Task Group on Reference Man", International Commission on Radiologic Protection. Pergammon Press: ICRP Publication No. 23, 1975. This is on order through EPA's Interlibrary Loan system (6/93).

Martha Waters at NIOSH recommended consulting "Ergonomic Design for People at Work" by Eastman Kodak. This is on order through EPA's Interlibrary Loan system (6/93).

Finally, Martha mentioned that LouAnn Beeckman of NIOSH, Morgantown was doing research on breathing rates. LouAnn is in the process of developing research protocol, and hopes to have a peer-review in September. They plan to collect data during the summer of '94, and will measure breathing rates during various tasks in

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[1] C. O. Jones, S. Gauld, J. F. Hurley, and A. M. Rickman. "Personal Differences in the Breathing Patterns and Volumes and Dust Intakes of Working Miners" Commission of the European Communities, Luxembourg: June, 1981.

[2] R. Smith and L. Stayner. "An Exploratory Assessment of the Risk of Lung Cancer Associated with Exposure to Diesel Exhaust Based on a Study in Rats". National Institute for Occupational Safety and Health/Division of Standards Development and Technology Transfer: n.d.

[3] R. Passmore and J.V. Durnin. "Human Energy Expenditures". *Physiol. Rev.* October 1955, Vol. 35. pp. 801-840.

[4] N. L. Jones, and E. J. Moran Campbell. *Clinical Exercise Testing*. W. B. Saunders Company: Philadelphia, PA 1982. Appendices A, B, and C.

various industries. There will be a FR notice for the peer review, and we could obtain a copy of the research protocol at that time by contacting NIOSH. LouAnn can be reached at (304) 291-4223.

CHRONIC, ACUTE AND PEAK EXPOSURES

Chronic Exposures. Chronic exposures are calculated using the LADD. Reference 1 (Exposure Factors Handbook) differentiates between carcinogenic and non-carcinogenic chronic exposures. The total lifetime (75 years) should be included in the carcinogenic chronic risk assessment. Whereas, only the total exposure duration should be included in calculating the non-carcinogenic Average Daily Dose. It's EPA policy in estimating chronic risk for occupational exposures to include the total lifetime in the calculation.

Chronic Exposure Health End-Points. Health end-points for chronic exposures are those which occur months or years after long-term exposure to a substance. Examples of chronic exposure health end-points are cancer, pneumoconioses caused by particulates (coal dust, asbestos, silica) and long-term kidney and liver damage.

Acute Exposures. Acute exposures are of concern due to immediate or short-term health effects from exposure to a given substance. In estimating potential acute exposures, the APDR excludes the LADD terms of total working years and total lifetime, and uses peak and short-term exposure data when available. For substances that have both acute and chronic effects, the risk should be assessed from both the LADD and the APDR and the lowest term used in defining use of the substance and appropriate protective equipment.

Acute Exposure Health End-Points. Health end-points of acute exposures are immediate or short-term effects caused by a single or short-term exposure. Examples include welding fume fever, dermatitis and immediate pulmonary irritation.

Peak Exposures.

Assessing peak exposures without sampling data in the work place is fraught with perils. The average exposure levels incorporate all peak levels. However, a given average exposure level may be a result of a constant exposure level, a series of intermittent exposures or a single peak exposure. Exposure to any substance is dependent on a variety of factors (e.g. work place controls, worker habits, process design, personal protective equipment, etc). When using models, professional judgement should be used to select input parameters that would provide a reasonable peak estimate, if possible.

Robert Harris, Advisory Panel for the Acrylonitrile Study Protocol, derived an equation (see the following page) to predicts the most likely maximum peak concentration based on 95% probability. The equation incorporates the geometric mean and geometric standard deviation of 8-hour sample results. Patricia Stewart, National Cancer Institute, stated that this equation is very poor at predicting peak exposures. This equation is not practical for CEB use because it depends on sample results. When CEB estimates potential exposures, sample results, for the most part, are not available.

Assumptions are identified in Appendix A.

3. For all other jobs which experience short-term excursions and only TWA (not peak) monitoring results are available, ceiling levels will be calculated from the geometric standard deviation as described by Harris (26) to determine if peaks exceeding 20 ppm were likely. This procedure predicts the maximum peak concentration as:

$$C_{pmax} = m_s(GM_s/m_s)^v((GSD_s)^{v0.5})^x$$

where:

m_s = the 8-hour arithmetic mean

GM_s = the geometric mean

$$v = \frac{\ln\ (t/(15\ \text{mins/peak}))}{\ln\ (t/(\#\text{min}/8\ \text{hours}))}$$

t = the time period to be assessed for peak exposures in minutes

GDS_s = The geometric standard deviation of the 8-hour samples

x = the number of standard deviations above the mean from a normal distribution. x = 2.96 when t = 1 month; 1.86 when t = 1 week and 1.67 when t = 1 day

3. Estimate of an Eight-hour Dose

The estimates developed thus far have been for exposures or air concentrations. Exposure, however, is only a surrogate for dose. There are many variables which affect dose, including distribution and metabolism of the

REFERENCES

1. U.S. Environmental Protection Agency, Office of Health & Environmental Assessment, Washington DC 24060, Exposure Factors Handbook, EPA/600/8-89/043, March 1990.

2. U.S. Environmental Protection Agency, Office of Pollution Prevention & Toxics, Chemical Engineering Branch, Preparation of Engineering Assessments, Volume 1: CEB Engineering Manual, February 1991.

3. U.S. Environmental Protection Agency, Office of Pollution Prevention & Toxics, Methods for Assessing Exposure to Chemical Substances, Volume 6: Methods for Assessing Occupational Exposure to Chemical Substances, August 1985.

4. Mark Pederson, "Low Dioxin Dry Chloranil, Revised RM1 Exposure Report, Processing Exposure Report - Manufacturing of Pigment Violet 23 with Chloranil, Inhalation and Dermal Exposure", November 1991.

5. U.S. Environmental Protection Agency, Office of Pollution Prevention & Toxics, Chemical Engineering Branch, "Engineering Support for Indoor Air Aerosol Spray Paint Cluster, Final Report", July 1992.

6. U.S. Environmental Protection Agency, Environmental Criteria & Assessment Office, "Chemical Mixtures Risk Assessment Guidelines", Course Material, Facilitator, Rita Schoeny, June, 1993.

7. Mary Ryan, "Comparison of EAB's and CEB's Methods for Assessing Dermal Exposure", April 1993.

8. Personal Communication between Robert Harris, Advisory Panel for Acrylonitrile Study Protocol and Patricia Stewart, National Cancer Institute, 1990.